Reproducing FUR: A Closer Look at Parametric Faithfulness

Project Proposal
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Abstract

Chain-of-thought (CoT) prompting has become a standard technique for eliciting reasoning from language models, yet it remains unclear whether the verbalized steps faithfully reflect the model's internal computations. Tutek et al. (2025, EMNLP) propose Faithfulness by Unlearning Reasoning steps (FUR), which unlearns information from individual CoT steps and observes the effect on the model's prediction. FUR shifts faithfulness evaluation from contextual perturbation to parametric intervention, but rests on a single instantiation of its loss function whose design choices are not systematically studied. We propose to (1) reproduce FUR on a 2×2 model-dataset subset, and (2) probe its loss function along two directions: varying the weight of the retain-side regularizer, and replacing the forget-side objective with alternatives.

Problem

FUR's central claim is that parametric intervention identifies more faithful CoTs than contextual baselines. This result, however, rests on a single instantiation of the loss $\mathcal{L} = \mathcal{L}_{\mathrm{NPO},\beta} + \mathcal{K}_{\mathrm{RT}}$. Two design choices in this loss are not systematically studied in the paper: the form of the weight of the retain-side regularizer (implicitly fixed at 1) and the forget-side term. It remains unclear whether FUR's reported numbers reflect a robust optimum or an arbitrary operating point under these choices.

Data and Model

We will use two MCQA datasets from the original paper: OpenBookQA and StrategyQA. For each, we will evaluate 50 instances (vs. 250 in the original) to fit our compute budget. The models will be two small open-weight instruction-tuned LMs used in the paper and official codebase: Phi-3-mini and LLaMA-3.2-3B. We also contacted the original authors and have obtained their raw output files for the ablation and final-result experiments, covering all model-dataset cells across multiple learning rates. These will serve as a per-cell reference for validating our reproduction at the per-instance level.

Approach

Reproduction. We will reproduce the core FUR pipeline on a 2×2 model-dataset subset rather than the full set of experiments. For each MCQA instance, we will generate a CoT with the paper's two-step prompting setup and segment it into sentence-level reasoning steps using NLTK. We will retain only steps with at least two SpaCy content words, then apply NPO+KL unlearning to the FF2 matrix of each Transformer block. This stage will use the official hyperparameters ($\beta = 0.1$, `npo_coeff` = `KL_coeff` = 1.0, five epochs) and the best learning rates reported by the original paper for each model-dataset cell. We will assess intervention validity using efficacy and specificity, and measure parametric faithfulness with FF-HARD and FF-SOFT under direct-answer prompting. These results will be compared with the Add-mistake contextual baseline, with Gemini-3-flash substituted for the unavailable GPT-4o-mini. The reproduction will deliver a result table reporting FUR scores against the Add-mistake baseline across the four model-dataset cells, efficacy and specificity sanity checks, and a case study on one representative instance, visualizing answer-probability shifts and the generated CoTs before and after unlearning."

Extension - FUR Loss-function probe. We will probe FUR's loss function along two complementary directions. First, we will modify the loss function of the original paper to $\mathcal{L} = \mathcal{L}_{\mathrm{NPO},\beta} + \lambda\mathcal{K}_{\mathrm{RT}}$ and vary λ (implicitly fixed at 1 in the original) to study the trade-off between forgetting strength, model specificity, and CoT faithfulness scores. Second, we will replace NPO with an alternative forget-side objective to test whether FUR's conclusions are method-specific. Candidates include SimNPO (Fan et al. 2024), which removes NPO's dependence on a reference model and is not considered by the original paper; Gradient Ascent (Jang et al. 2023), the simplest unlearning baseline. Other gradient-based variants (e.g., Gradient Difference) are also under consideration. The final selection will be made after a feasibility check, and we expect to report one alternative.